

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Hashini S / 192524035 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Hashini S

1. Constraint Satisfaction Problem (Backtracking Search)

Problem

Assign three doctors (**D1**, **D2**, **D3**) to three shifts:

- Morning (M)
- Afternoon (A)
- Night (N)

Constraints

1. D1 cannot work Night.
2. D2 must work before D3.
 - Morning < Afternoon < Night
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. Every doctor gets exactly one shift.

Step 1: Variables and Domains

Doctor	Possible Shifts
D1	Morning, Afternoon
D2	Morning, Afternoon, Night
D3	Afternoon, Night

Step 2: Backtracking Search

Initial State

No assignments.

D1 = ?

D2 = ?

D3 = ?

Assign D1 = Morning

Current Assignment

D1 = Morning

Constraint Check

- D1 $\neq$ Night ✓
- No duplicate shifts ✓

Continue.

Assign D2 = Afternoon

Current Assignment

D1 = Morning

D2 = Afternoon

Constraint Check

- Unique shifts ✓
- D2 before D3 (not assigned yet) ✓

Continue.

Assign D3 = Night

Current Assignment

D1 = Morning

D2 = Afternoon

D3 = Night

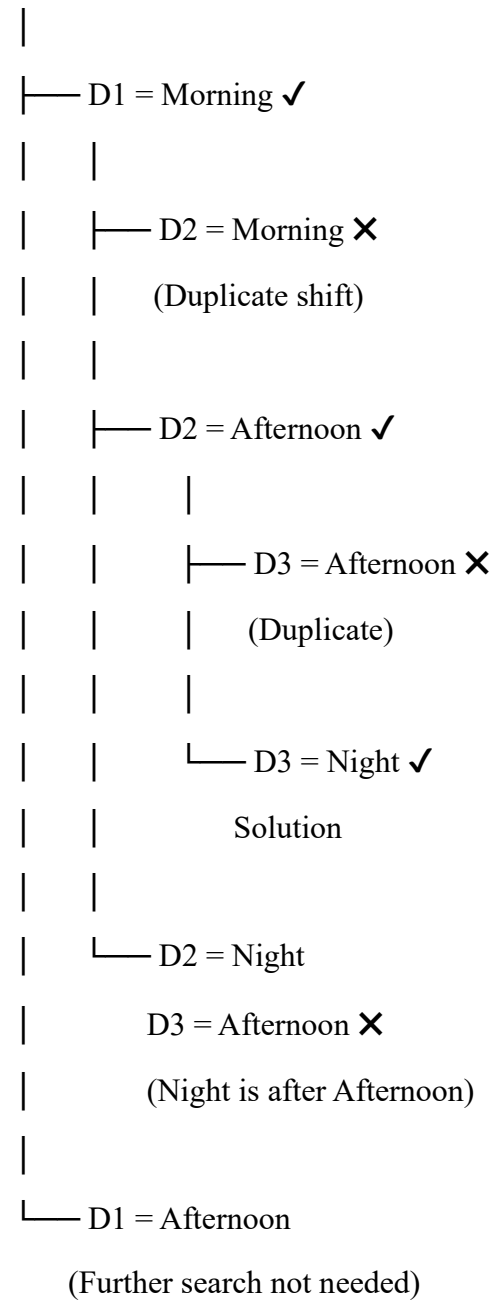
Constraint Check

- D3 $\neq$ Morning ✓
- D2 before D3
 - Afternoon < Night ✓
- One doctor per shift ✓
- All assigned ✓

Solution found.

Search Tree

Start



Final Valid Schedule

Shift	Doctor
Morning	D1

Afternoon	D2
Night	D3

2. Robot Grid Problem (BFS Search)

Given Grid (5 × 5)

We'll use the following grid.

S . X . .

. . X . .

.

X X . X .

. . . . G

Legend

- S = Start
- G = Goal
- X = Obstacle
- . = Free cell

Coordinates

(0,0) S

(4,4) G

Manhattan Distance Formula

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

Goal = (4,4)

Initial Queue

Queue = [(0,0)]

Visited = {(0,0)}

Step-by-Step BFS Expansion

Step 1

Expand

(0,0)

Neighbours

(1,0)

(0,1)

Queue

[(1,0),(0,1)]

Step 2

Expand

(1,0)

Add

(2,0)

(1,1)

Queue

[(0,1),(2,0),(1,1)]

Step 3

Expand

(0,1)

New node

None

Queue

[(2,0),(1,1)]

Step 4

Expand

(2,0)

Add

(2,1)

Queue

[(1,1),(2,1)]

Step 5

Expand

(1,1)

No new nodes.

Queue

[(2,1)]

Step 6

Expand

(2,1)

Add

(2,2)

Queue

[(2,2)]

Step 7

Expand

(2,2)

Add

(3,2)

(2,3)

Queue

[(3,2),(2,3)]

Step 8

Expand

(3,2)

Add

(4,2)

Queue

[(2,3),(4,2)]

Step 9

Expand

(2,3)

Add

(1,3)

(2,4)

Queue

[(4,2),(1,3),(2,4)]

Step 10

Expand

(4,2)

Add

(4,3)

(4,1)

Queue

[(1,3),(2,4),(4,3),(4,1)]

Step 11

Expand

(1,3)

Add

(0,3)

(1,4)

Queue

[(2,4),(4,3),(4,1),(0,3),(1,4)]

Step 12

Expand

(2,4)

No new nodes.

Queue

[(4,3),(4,1),(0,3),(1,4)]

Step 13

Expand

(4,3)

Add

(4,4) Goal

Queue

[(4,1),(0,3),(1,4),(4,4)]

Goal reached.

Optimal Path

(0,0)

↓

(1,0)

↓

(2,0)

→

(2,1)

→

(2,2)

↓

(3,2)

↓

(4,2)

→

(4,3)

→

(4,4)

Manhattan Distances Along the Path

Node h(n)

(0,0) 8

(1,0) 7

(2,0) 6

(2,1) 5

(2,2) 4

Node $h(n)$

(3,2) 3

(4,2) 2

(4,3) 1

(4,4) 0

Total Cost

Each move costs 1.

Moves:

8

Optimal Cost = 8

3. Autonomous Rescue Robot

(a) Constraint Analysis

Constraint	Effect on Decision Making
Move only Up, Down, Left, Right	Limits the robot to four possible actions at each step.
Obstacles cannot be crossed	The robot must avoid blocked cells and search for alternate paths.
Limited sensing (adjacent cells only)	The robot cannot see the whole map initially and must explore gradually.
Move cost = 1	Encourages finding shorter paths.
Risky zone cost = +2	The robot should avoid risky areas unless necessary, balancing distance and safety.
Dynamic knowledge update	As new cells are observed, the robot updates its internal map and replans if needed.

(b) Solution Approach – Uniform Cost Search (UCS) with Online Search

Step 1

Start from S with cost 0.

Step 2

Sense adjacent cells.

Step 3

Identify free cells, obstacles, and risky zones.

Step 4

Insert reachable cells into a priority queue ordered by cumulative cost.

Step 5

Expand the node with the lowest total cost.

Step 6

Update the map whenever new information is discovered.

Step 7

Repeat until the survivor (**G**) is reached.

Step 8

Return the minimum-cost path.

(c) AI Concepts Applied**Intelligent Agent**

- Perceives the environment through sensors.
- Chooses actions to move safely.
- Continuously updates its knowledge.

Rationality

- Always selects the action with the lowest expected total cost.
- Avoids risky zones unless they provide the best overall route.

Environment Type

Property	Type
Observability	Partially Observable

Determinism	Deterministic
Dynamics	Dynamic (knowledge changes during exploration)
Agent Type	Single Agent
State Space	Discrete

(d) Justification of the Chosen Approach

Uniform Cost Search combined with an Online Search Agent is the most suitable approach because:

- It guarantees the **minimum-cost path** when movement costs vary.
- It naturally accounts for **additional risk costs**.
- It can **replan dynamically** as the robot discovers new obstacles or routes.
- It efficiently balances **path length** and **safety**, making it well suited for disaster rescue scenarios where the environment is only partially known.

Conclusion

By using Uniform Cost Search with online updates, the rescue robot can safely navigate around obstacles, minimize travel and risk costs, adapt to newly discovered information, and reliably reach the survivor.